11. KPI Definitions and Calculations

Verified baseline KPIs

KPI | Calculation | Raw-file result | Interpretation

Total records | COUNT(vehicle_id) | 100,000 | Rows in supplied file; not sales volume

Distinct vehicle IDs | COUNTD(vehicle_id) | 100,000 | Unique IDs in raw file

Manufacturers | COUNTD(manufacturer) | 200 | Distinct labels

Categories | COUNTD(category) | 15 | Distinct vehicle-category labels

Regions | COUNTD(market_region) | 7 | Distinct region labels

Mean price | AVG(price_usd) | $65,010.46 | Sensitive to high-priced vehicles

Median price | MEDIAN(price_usd) | $42,808.69 | Robust central estimate

Mean range | AVG(range_km) | 324.31 km | Dataset average, not guaranteed real-world range

Median range | MEDIAN(range_km) | 338.30 km | Dataset median

Mean efficiency | AVG(efficiency_wh_per_km) | 250.57 Wh/km | Lower values indicate lower energy per distance if units are consistent

Mean customer rating | AVG(customer_rating) | 4.08 | Only {int(df.customer_rating.notna().sum()):,} non-null records

Mean CO■ saved | AVG(co2_saved_tons_per_year) | 3.69 tons/year | Methodology/baseline not supplied

Tableau calculations

- Record count: `COUNT([vehicle_id])`
- Distinct manufacturer count: `COUNTD([manufacturer])`
- Average price: `AVG([price_usd])`
- Median price: `MEDIAN([price_usd])`
- Average range: `AVG([range_km])`
- Average efficiency: `AVG([efficiency_wh_per_km])`

KPI governance

Do not label `SUM([annual_sales_units])` as global market sales without verifying that each row's sales measure is additive, non-overlapping, and represents the stated period and geography. The dataset contains one row per unique `vehicle_id`, but the scope of the annual sales field is not documented.